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Overall Algorithm

WelcomePage

If (**Login** Button is pressed) {Go to **LoginPage**}

LoginPage

If (**Login** Button is clicked) {Direct to main dashboard}

If (**RequestHelp** Button is pressed) {Go to **KnowledgeBasePage**}

AdminDashboardPage

If (MessageTile is clicked) {Show Message}

//FUNCTIONBAR (ADMIN)

If (FunctionBar.HRResource is clicked) {Go to **HRPage**}

If (FunctionBar.EmployeeManagement is clicked) {Go to **EmployeeManagementPage**}

If (FunctionBar.KnowledgeBase is clicked) {Go to **KnowledgeBasePage**}

If (FunctionBar.SendMessage is clicked) {Send announcements/notices to all employees}

ManagerDashboardPage

If (MessageTile is clicked) {Show Message}

//FUNCTIONBAR (MANAGER)

If (FunctionBar.PerfEvaluation is clicked) {Go to **PerformanceEvaluationPage**}

If (FunctionBar.EmployeeInfo is clicked) {Go to **EmployeeInfoDBPage**}

If (FunctionBar.KnowledgeBase is clicked) {Go to **KnowledgeBasePage**}

If (FunctionBar.SendMessage is clicked) {Send announcements/notices to all employees}

EmployeePage

If (MessageTile is clicked) {Shows messages sent by Admin/Managers}

//FUNCTIONBAR (EMPLOYEE)

If (FunctionBar.KnowledgeBase is clicked) {Go to **KnowledgeBasePage**}

KnowledgeBasePage

If (KnowledgeBase Tile is clicked) {Shows Knowledgebase article}

If (**ReturnToDashboard** Button is pressed) {Return to main dashboard}

HRPage

If (**BudgetSalary** Button is clicked) {Go to **MoneyPage**}

If (**Attendance** Button is clicked) {Go to **AttendancePage**}
If (**PerformanceEvaluation** Button is clicked) {Go to **PerformanceEvaluationPage**}
If (**ViewDB** Button is clicked) {Go to **PerformanceEvaluationDBPage**}
//FUNCTIONBAR (ADMIN) HERE

EmployeeManagementPage

If (**CreateAccount** Button is clicked) {Open **AccountCreation Popup Window**}
If (**AddEmployee** Button is clicked) {Open **AddEmployee Popup Window**}
If (**ViewDB** Button is clicked) {Go to **EmployeeInfoDBPage**}

//FUNCTIONBAR (ADMIN) HERE

EmployeeInfoDBPage

Display information from EmployeesTable (Database Table)
If (ReturnToEmployeeManagementPage Button is clicked) {Go to
EmployeeManagementPage}

MoneyPage

If (**ModifyBudget** button is pressed) {Open **Budget Popup Window**}
If (EmployeeID is selected from dropdown)
{
 Display Employee's Year at the company and their CurrentSalary
 If (**ViewDB** Button is pressed) {Go to **EmployeeInfoDBPage**}
 If (**BonusCalc** Button is pressed) {Open **BonusCalculation Popup Window**}
}
If (**ReturnToHR** Button is pressed) {Go to **HRPage**}

AttendancePage

If (Employee is marked present) {Add to Employee Attendance Record (Presences)}
Else {Add to Employee Attendance Record (Absences)}
If (**ReturnToHR** button is pressed) {Go to **HRPage**}

PerformanceEvaluationPage

If (EmployeeID is selected)
{
 Display Name, Attendance Record
 Display "Evaluation" Textfield
 If (**Store** button is pressed) {Add the performance evaluation content in DB}

}

If (**ReturnToDashoard** button is pressed) {Return to main dashboard}

PerformanceEvaluationDBPage

Display all performance evaluations from PerformanceEvaluationsTable (Database Table)

Normalized Tables

UsersTable

Description: For all users (Admins/Managers/Employees)

UserNumber (Primary Key, Int)	Username (String)	Password (String)	Role (String)
1	JohnDoe	123ABC	Admin
2	JaneDoe	123ABC	Manager
3	JoneDoe	123ABC	Employee

EmployeesTable

Description: Specific to employees ONLY

EmployeeID (Primary Key, Int)	UserNumber (Int)	First Name (String)	Last Name (String)	DepartmentID (Int)	DateOfBirth (String)	Salary (Double)
1	3	John	Doe	1	01-31-2001	50000.00

DepartmentsTable

Description: All departments in the company

DepartmentID (Primary Key, Int)	DepartmentName (String)
1	Human Resources
2	IT
3	Sales

MessagesTable

Description: Company-wide messages posted by the admin/manager

MessageID (Primary Key, Int)	SenderUserNa me (String)	MessageTitle (String)	MessageConte nt (String)	SentDate (String)
1	JohnDoe	Test1	This is a test.	01-31-2024
2	JaneDoe	Test2	This is a test.	01-31-2024

KnowledgeBaseTable

Description: All knowledge base articles posted by admins

ArticleID (Primary Key, Int)	Publisher (String)	Title (String)	Content (String)	PublishDate (String)
1	JohnDoe	Article1	Content here.	01-31-2024
2	JohnDoe	Article2	Content here.	01-31-2024

AttendanceTable

Description: Attendance Record for all employees

EmployeeID (Primary Key, Int)	PresentCount (Int)	AbsentCount (Int)	AttendanceYear (Primary Key, Int)
3	185	25	2024

PerformanceEvaluationTable

Description: All performance evaluations done by admins/managers

EvaluationID (Primary Key, Int)	EmployeeID (Int)	Evaluator (String)	DepartmentID (Int)	EvaluationDate (String)	EvaluationContent (String)
---------------------------------------	---------------------	-----------------------	-----------------------	----------------------------	-------------------------------

1	3	JohnDoe	1	01-31-2024	Testing 123
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BudgetTable

Description: All budget entries

BudgetID (Primary Key, Int)	Departme ntID (Int)	BudgetCa tegory (String)	Allocated Amount (Double)	AmountS pent (Double)	AmountR emaining (Double)	BudgetDa te (String)
1	2	Equipmen t	20000.00	999.99	19000.01	01-31-202 4

Class Descriptions

Classes	Purpose
Welcome	Direct user to log-in.
LogIn	Log-In Functionality and redirect to appropriate pages based on access level.
AdminDashboard	Dashboard Page for admins to access HR resources, budget management, and employee management.
ManagerDashboard	Dashboard Page for managers to perform performance evaluations and access department-specific employee data.
EmployeeDashboard	Employee Dashboard Page with limited ability to receive messages and view knowledge base only.
KnowledgeBase	Access to a guide to the app & content posted by the admin/management such as instructions.
HRResources	Human Resources functionalities such as payrolls/bonus/attendance/performance evaluation.
EmployeeManagement	Ability to view employee information or create an account for employees.
EmployeeInfoDBDisplay	Shows the Employee Information in a database.
Money	Handles budget allocation, bonus calculations, and tracks budget categories.
Attendance	Ability to take attendance of employees for admins
PerformanceEvaluation	Being able to perform a performance evaluation for employees under their department (Managers) or all employees (Admins).
PerformanceEvaluationDB	Shows all performance evaluations.

ErrorPopup	Used for any error popups
MessagePopup	Used for any messages/knowledge base viewing
SendMessagePopup	Used for sending messages to employees
ModifyBudgetPopup	Used to calculate all budget-related queries such as total amount, remaining budget, and percentage allocation
BonusCalculationPopup	Used for calculating budgets based on staff salary and the rate of the bonus

UML & Diagram

Welcome

-welcomeMessage: String
-//GUI Components

+Welcome()
+actionPerformed(ActionEvent e): void

Login

-//GUI Components

+Login()
+checkAccess(): void
+actionPerformed(ActionEvent e): void

AdminDashboard

-username: String
-userID: Int
-//GUI Components

+AdminDashboard()
+actionPerformed(ActionEvent e): void

ManagerDashboard

-username: String
-userID: Int
-//GUI Components

+ManagerDashboard()
+actionPerformed(ActionEvent e): void

EmployeeDashboard

-username: string
-userID: Int
-employeeName: String
-employeeID: Int
-//GUI Components

+EmployeeDashboard() +actionPerformed(ActionEvent e): void

KnowledgeBase

-articleList: ArrayList -//GUI Components
--

+KnowledgeBase() +displayArticle(int articleID): void +addArticle(String title, String content): void +actionPerformed(ActionEvent e): void
--

HRResources

-//GUI Components

+HRResources() +actionPerformed(ActionEvent e): void

EmployeeManagement

-//GUI Components

+EmployeeManagement() +createUser(string userName; string password; string role): void +addEmployee(int userNumber; string firstName; string lastName; int departmentID; string dateOfBirth; double salary): void +actionPerformed(ActionEvent e): void
--

EmployeeInfoDB

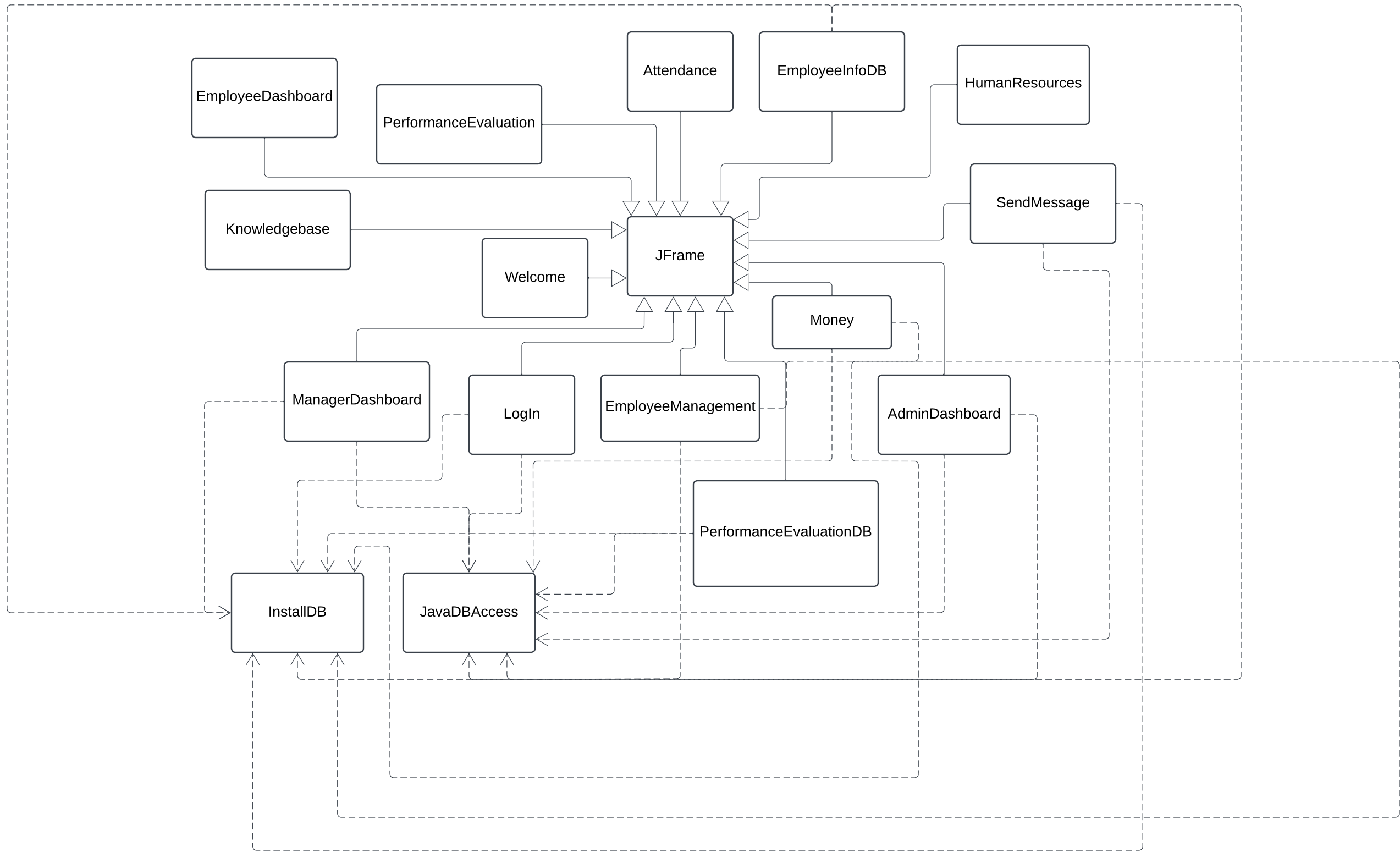
-employeeData: ArrayList -//GUI Components

+EmployeeInfoDBDisplay() +actionPerformed(ActionEvent e): void

Money

-budgetData: ArrayList -bonusAmount: double
--

-//GUI Components
+Money() +allocateBudget(int departmentID, double amount): void +calculateBonus(double salary, double bonusRate): void +actionPerformed(ActionEvent e): void
Attendance
-attendanceRecords: ArrayList -//GUI Component
+Attendance() +markAttendance(int employeeID, boolean isPresent): void +actionPerformed(ActionEvent e): void
PerformanceEvaluation
-evaluationID: int -employeeID: int -content: String -evaluator: String -//GUI Components
+PerformanceEvaluation() +PerformanceEvaluation(evaluationID, int employeeID, String content, String evaluator) +addEvaluation(String content, String evaluator): void +actionPerformed(ActionEvent e): void
PerformanceEvaluationDB
-evaluationData: ArrayList -//GUI Components
+PerformanceEvaluationDB() +searchEvaluation(int employeeID): void +actionPerformed(ActionEvent e): void



GUI Draft



Welcome to Name OA System

Log-In → Log-In Page

Log - In

UserID:

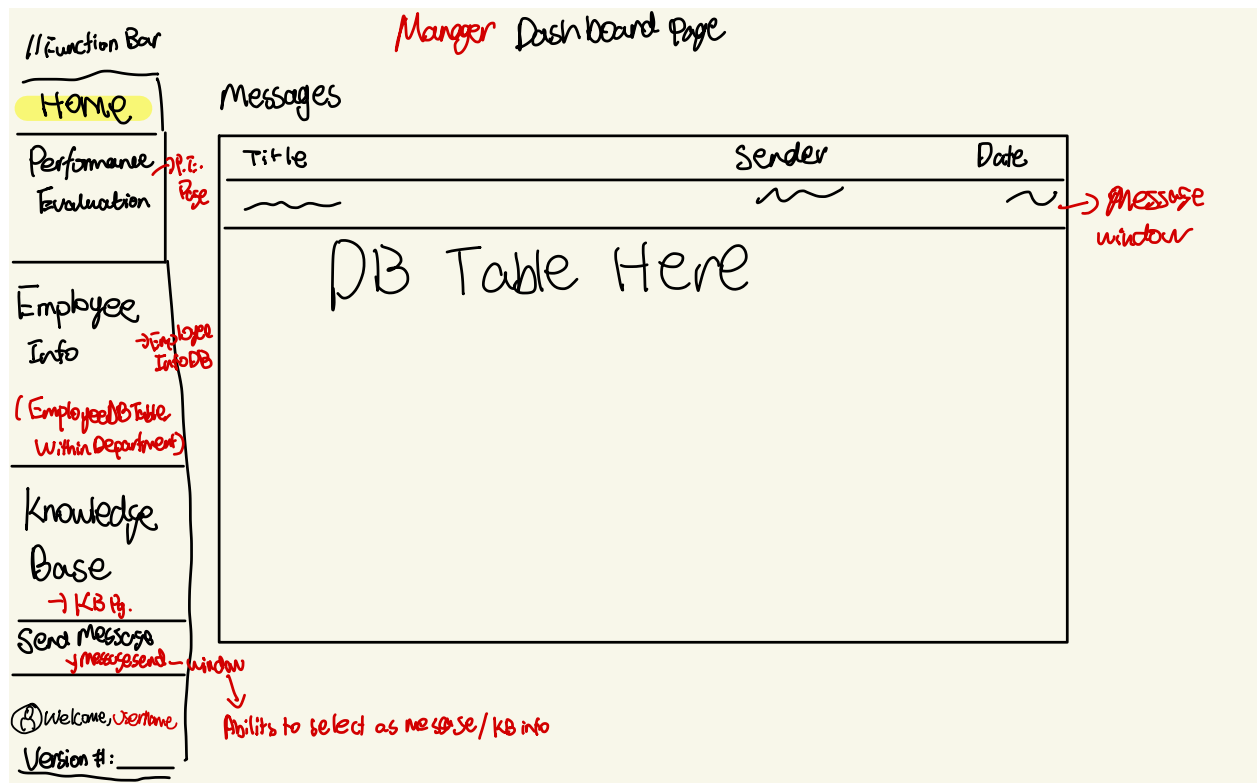
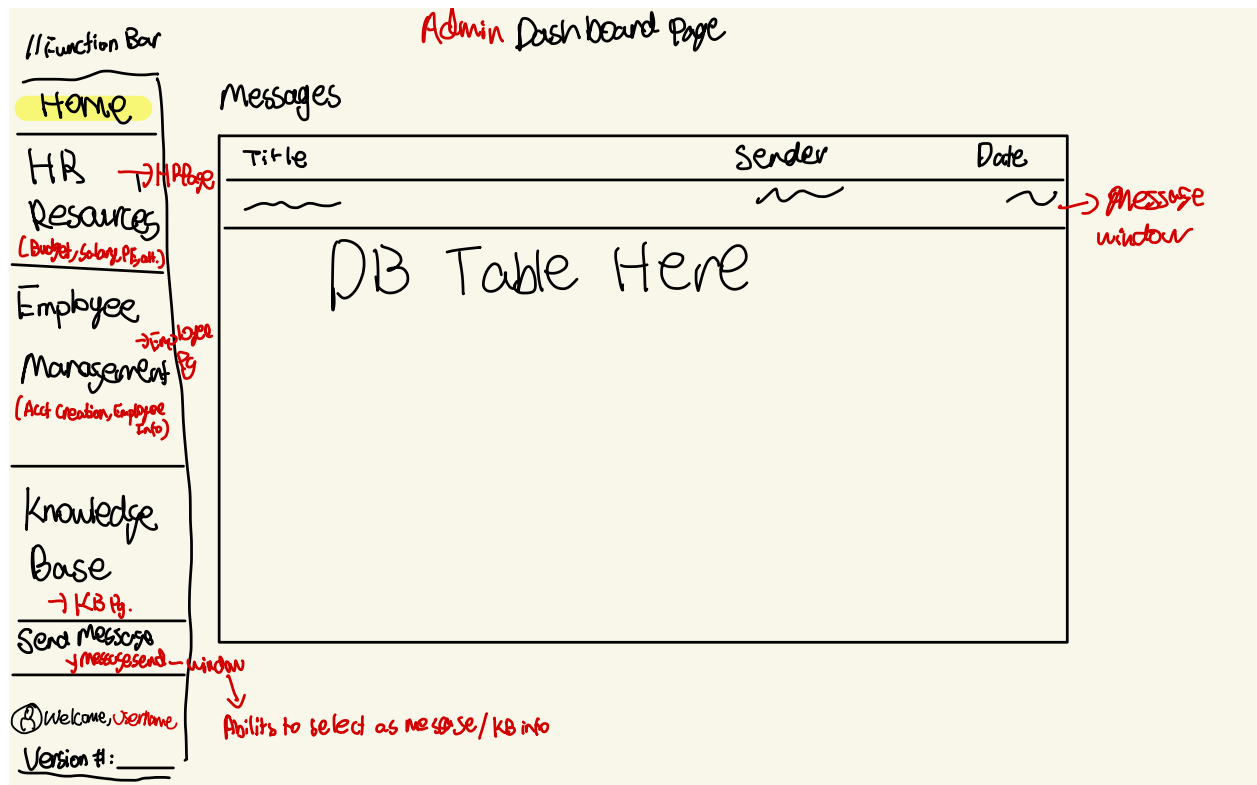
Password:

Error window
if Error

Log-In

→ DashboardPage/Employee View

Request Help? → KB Page



//Function Bar

Employee Page

Messages

Home

Knowledge Base

JKBB.

Ⓟ Welcome, Username

Version #: _____

Title	Sender	Date
DB Table Here		

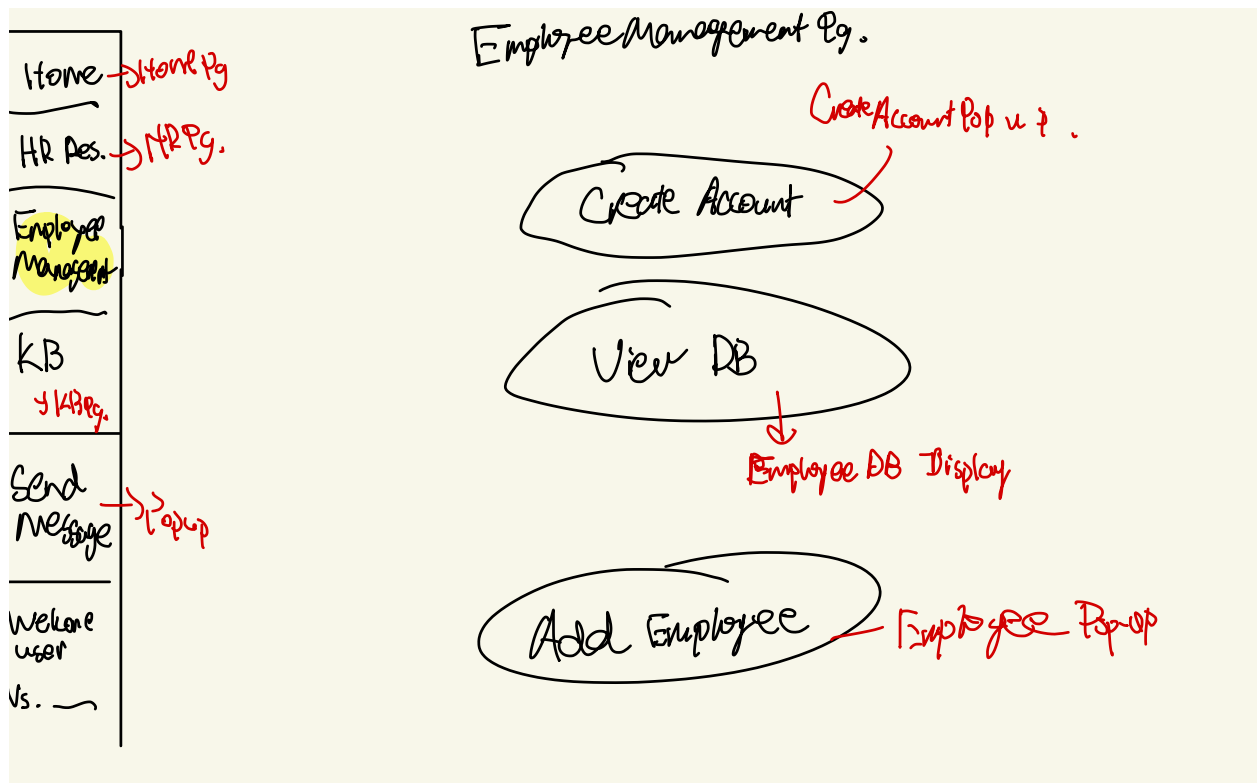
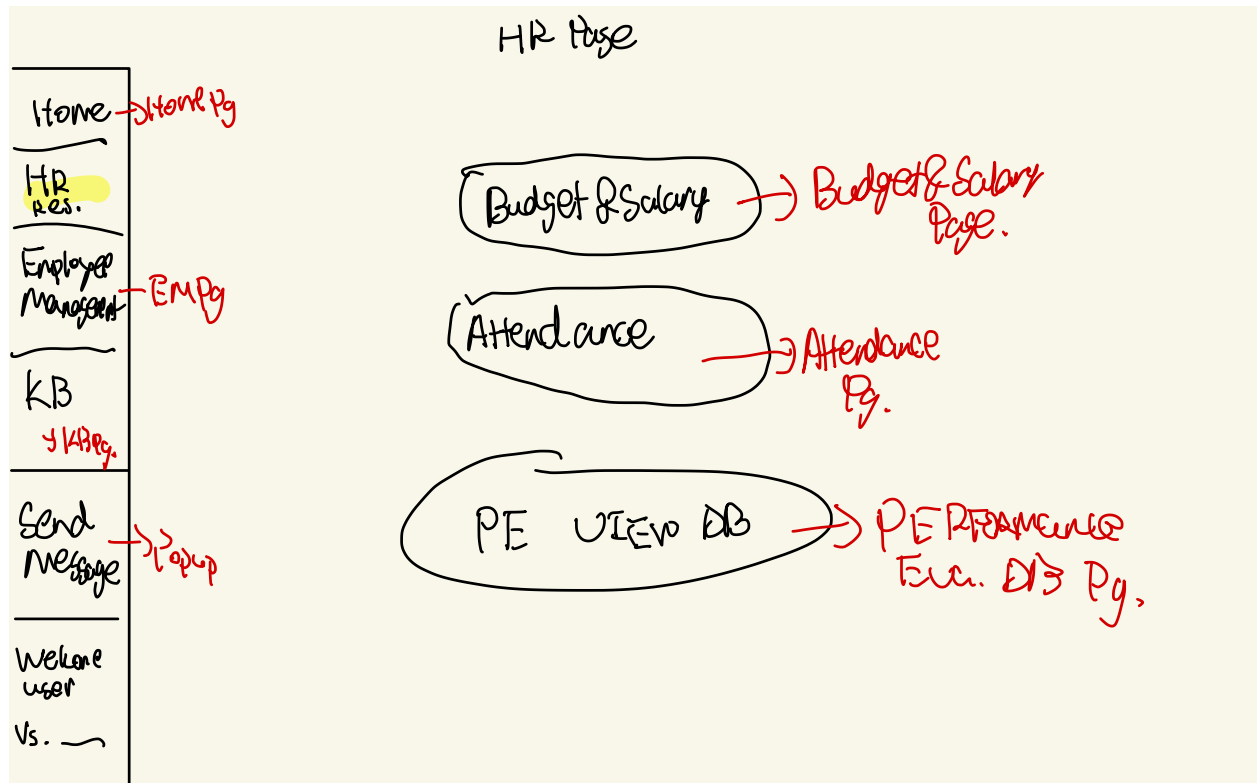
Message Pop-up Window

Knowledge Base Pg.

Message Pop-up Window

Title	Published By	Date Published
-		

Return to Home



EmployeeDB

Name	VerID	ret. ID	Age	DOB	Salary	Bonus

Return To Employee Mgr. Employee Manager's

MoneyPg.

Budget Allocation

Dept.

Graph Here

Modify

Employee Salary & Bonus

Name ▾

Years at Company:

Current Salary: \$

Bonus Calc Bonus Popup

View DB Employee DB

Budget Popup.

Return To HR HR Pg.

Algorithm
Calc.

Attendance Pg.

Name	Date	Attendance
~~~~~	~~~~~	DropBox ▾

[Return To HR](#) ~~HR Pg.~~

Name	Date	Attendance
<u>                    </u>	<u>                    </u>	Dropbox ▾

Return To HR ~~HR Pg.~~

# Performance Evaluation

Employee Performance Evaluation

Name:

Attendance Record:  Absent,  Present

Evaluation:

[Return to Dashboard](#) Dashboard Pg.

Employee Performance Evaluation

Name:

Attendance Record:  Absent,  Present

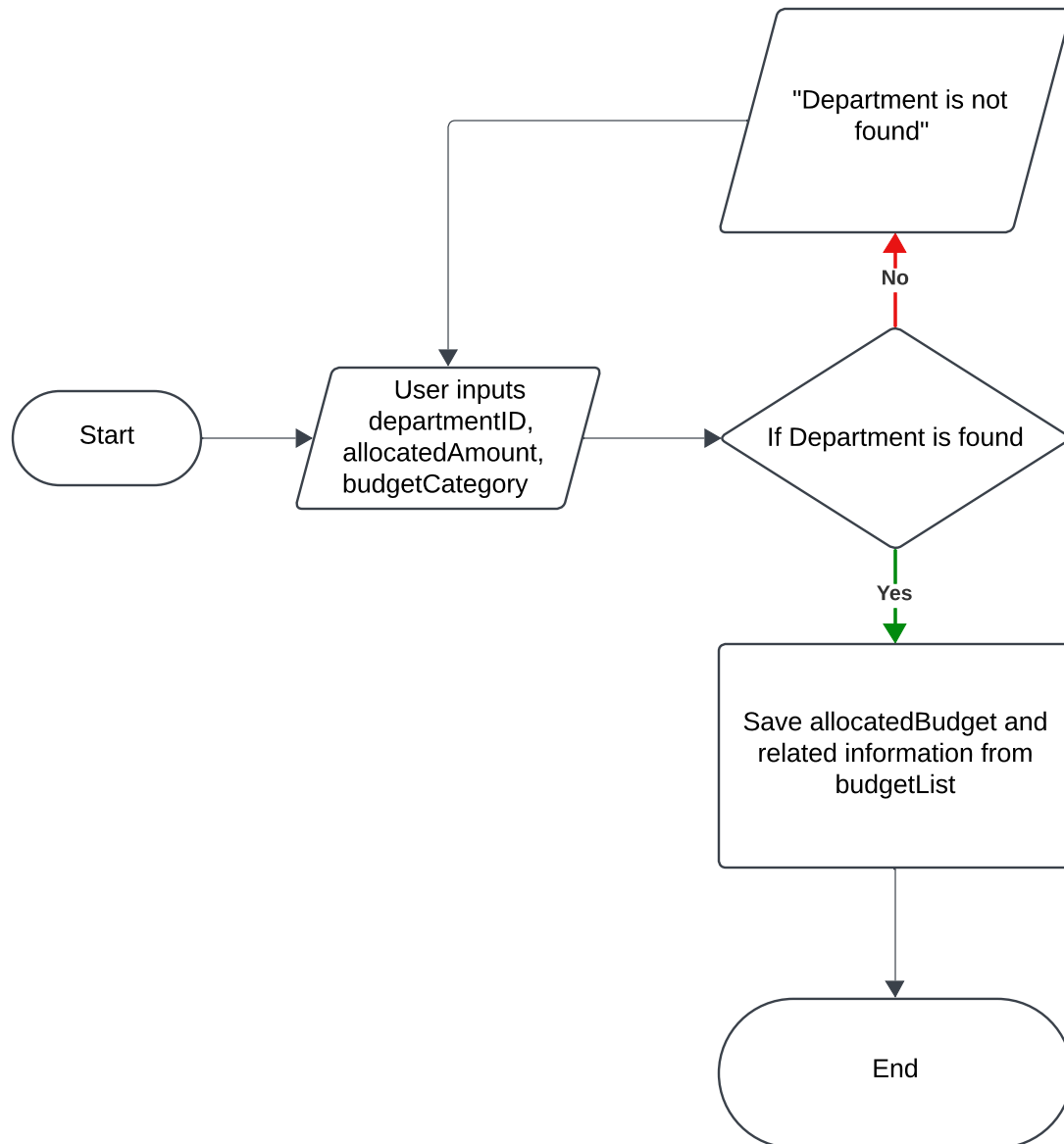
Evaluation:

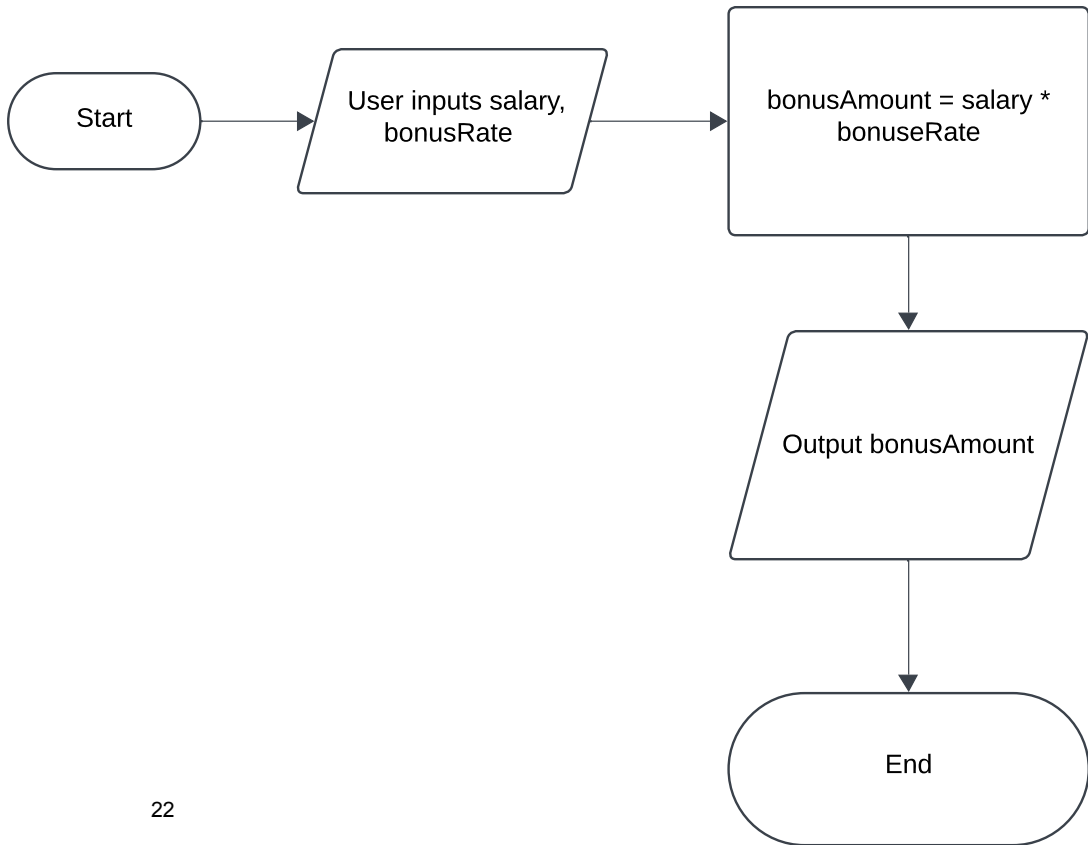
store

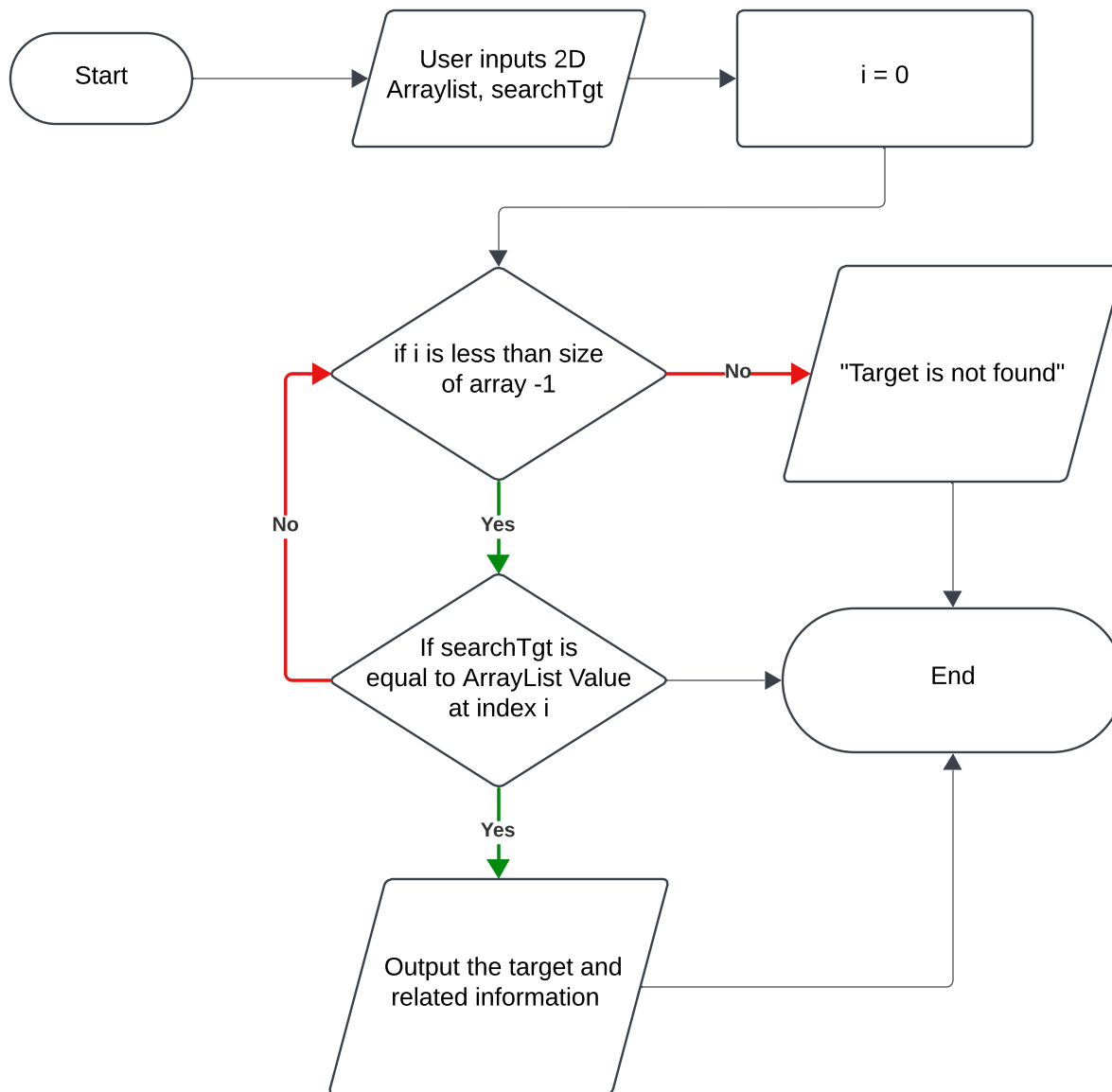
Return to Dashboard Dashboard Pg.

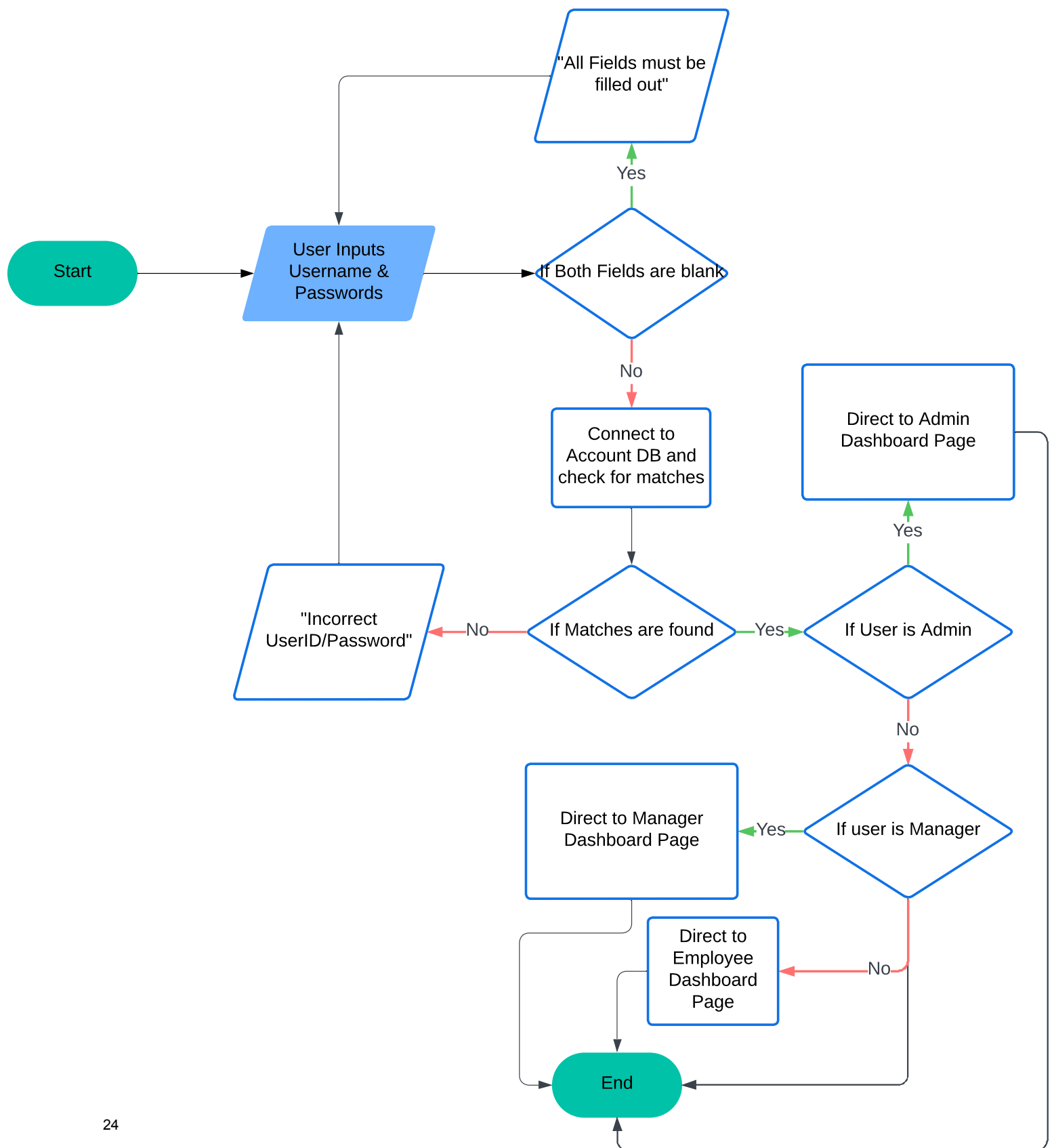
Error Popup    X  Title:  Text Here	Message Popup    Title:  Sender:  Date:  Content:	SendMessage Popup    X  Title:  Content:  Cancel	Modify Budget Popup    New Budget:  Budget Category:  Add
Bonus Calc Popup    Current Salary:  Bonus %:  Calc			

# Computational Flowcharts

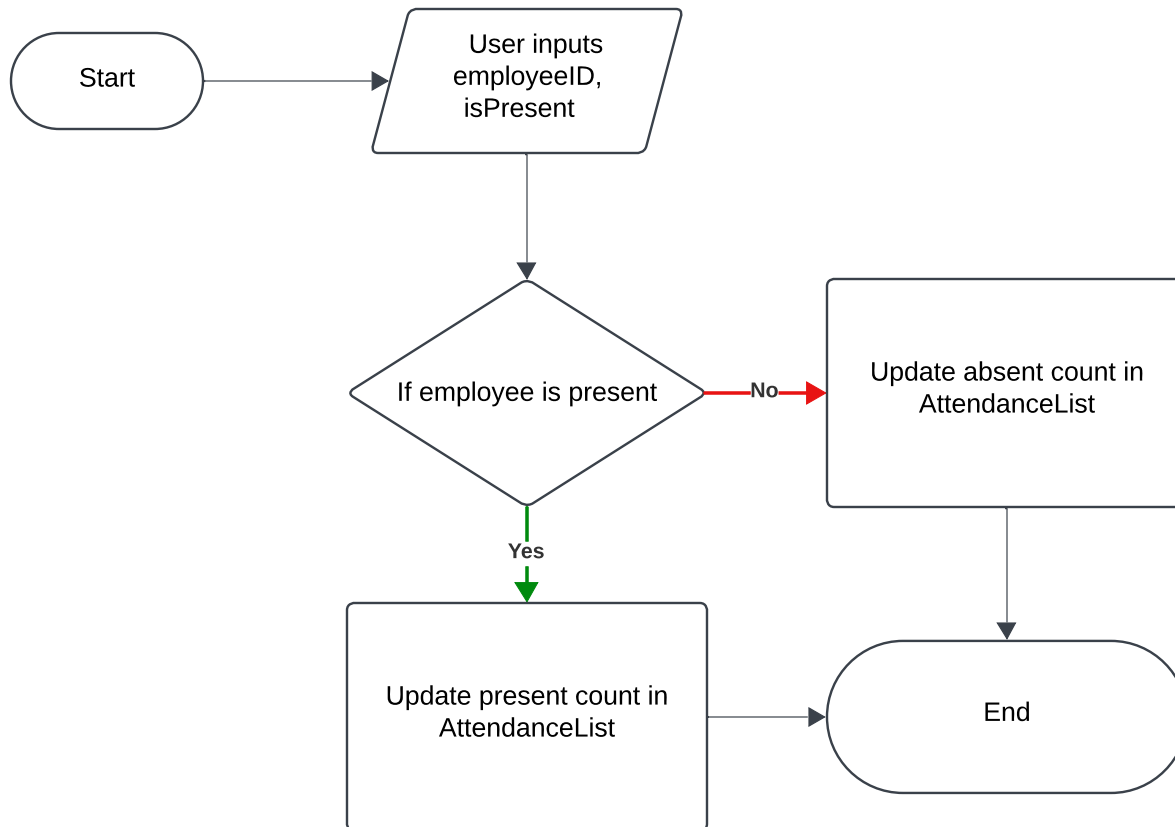


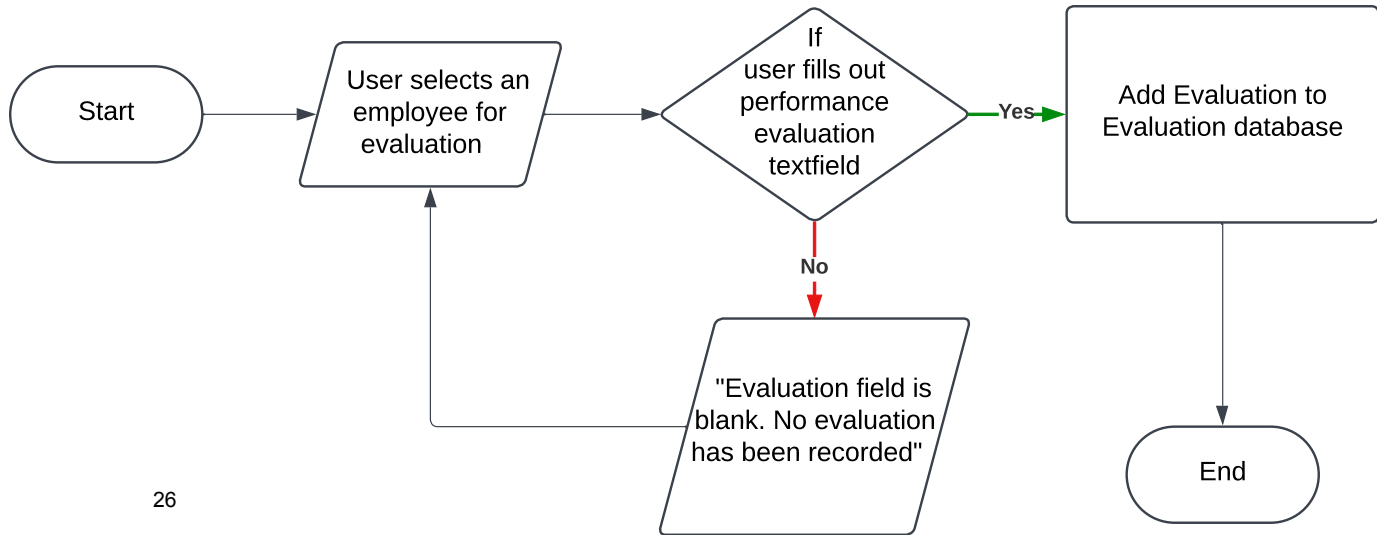












# Data Structure Tables

Variable Name	Data Structure	Purpose
UserList	2D ArrayList	Stores user data such as username, password, role from UserAccountsTable (Database)
MessageList	2D ArrayList	Stores messages with sender, date, title, and content from the MessagesTable (Database)
PerformanceEvaluationList	2D ArrayList	Stores employee performance evaluation including evaluationID, content, and evaluator from PerformanceEvaluationTable (Database)
AttendanceList	2D ArrayList	Stores the EmployeeID, PresentCount, AbsentCount, and Attendance Year. Obtained from AttendanceTable (Database)
EmployeeList	2D ArrayList	Stores all employee information such as names, date of birth, user number, salary, department, and employee ID from EmployeesTable (Database)
BudgetList	2D ArrayList	Stores budget data including allocated budget, spent budget, remaining budget, and budget category. Used for visualization of pie-chart. Obtained from BudgetTable (Database)
KnowledgeBaseList	2D ArrayList	Stores knowledge base articles including ArticleID, Title, publisher, content, and Date Published from KnowledgeBaseTable (Database)
DepartmentList	Arraylist	Stores all departments from DepartmentTable (Database)

# Test Plans

Success Criteria	Testing Plan
Allow login for all three levels of users which are Admin, Manager, and Employee	• White Box Testing     • Test if users are able to successfully log-in and direct to the correct pages
Admins can insert/update/delete all employees information	• User Acceptance Testing     • Admins create/read/update/delete accounts for employees.
Managers can view employee information for their department only	• User Acceptance Testing     • Attempt to view employees' information lower than the manager's level to ensure restricted access. Make sure the employeeDB is correct.
Admins can store performance evaluations for all employees and managers can for in-department employees	• Black Box Testing     • Add evaluations for employees under admin and manager accounts. Ensure that the data is saved in the PerformanceList and visible only to Admins/Managers.
Salary and bonus computation is correct based on the model inputted by the user	• White Box Testing     • Input different salary and bonusRate value to verify the computation is correct ($\text{salary} * \text{bonusRate}$).
All employees are viewable and searchable in a table.	• Black Box Testing     • Search for specific employees by name or employeeID and make sure that it is corresponding to the correct employee.
Attendance System for Admins to mark each employee for absences and presences	• White Box Testing     • Test marking an employee as present or absent and make sure that either count updates in the database.
Message Board where Admins and Managers could post announcements and notices and employees can view the messages.	• User Acceptance Testing     • Post messages as an admin/manager and verify that the employees are able to view the message. Test that only admins/managers can post messages.
Budget allocations and related information are displayed in a pie-chart format viewable by Admins.	• User Acceptance Testing     • Test if the data from budgetList shows through the pie-chart in the correct department/category.

Minimalistic and Simplistic Design in interface where main features are highlighted.	• User Acceptance Testing     • The design is intuitive with clarity and intuitive navigation.
Access and Entry will be restricted by the system based on the role of the user	• White Box Testing     • Attempt to access unauthorized pages such as admin pages from a manager/employee account. Make sure that level-restricted features are unavailable to lower levels.